Intermittency, period adding and bifurcation curves in a language model's own inference map

Authorship to be decided by the human researcher

Private working draft, 12 October 2026

Status of this draft. Results without a marker are numerically validated in float64 (evidence rung 1 or 2 in the sense of Section 14); **[Planned:** red] marks planned work and **[In progress:** orange] marks running work. Nothing here is a computer-assisted proof yet.

Abstract

A frozen language model defines its own iterated map when its expected next-token embedding is fed back as the next input. For Qwen2.5-0.5B-Instruct with a fixed chat prefix, a one-token window, sampling temperature T and an activation-steering strength α, this map of $\mathbb{R}^{896}$ is smooth, and we treat it with the tools of numerical bifurcation theory. Its $(\log_{10} T, \alpha)$ plane contains deterministic chaos with embedded periodic windows. A family of windows of periods $6, 8, 10, \ldots$ (up to 32 in the maps) reads as text ". This" followed by k pairs "is X". The windows accumulate on the fold of the 2-cycle "is it" as $\alpha_N - \alpha_c \propto N^{-2}$, where N counts the laps of the "is it" loop. We compute the saddle-node normal form of that fold from derivatives of the network. Without any fitted parameter, it predicts bottleneck passages of 7 to 144 iterates within one iterate. Natural trajectories re-enter the bottleneck near its middle, which fixes the prefactor of the window law. This is type-I intermittency (Pomeau–Manneville) inside the inference computation of a pretrained language model. We continue fold and flip curves of the periodic orbits with float64 test functions. In a higher-temperature region we find a fold curve with two cusps whose normal-form coefficient changes sign, the codimension-two organization of shrimp-shaped windows. In a closed region of the map a period-3 text loop doubles through periods 6, 12 and 24 toward a chaotic core; along a cut, float64 flip parameters are resolved through period 24 so far, with a first finite-order interval ratio of 2.84 **[In progress:** further flips and ratios]. Replacing the expected embedding by averages of m sampled embeddings shows how much of the mechanism survives sampling. Qwen2.5-1.5B shows chaos with embedded windows as well. A recurrent-depth model (Huginn-0125) gives a second system, in which a Neimark–Sacker bifurcation predicted from the spectrum of the native fixed point appears where predicted. **[Planned:** replication across prompts, steering directions and model sizes; misaligned behaviour (hostility) as an attractor of sampled self-dialogue: capture threshold, hysteresis and early warning.]

1 Introduction

Repeated inference with a fixed language model is an iterated map: the model's output becomes its next input. Recent work has measured chaos-like sensitivity in language-model generation [1, 2], temperature-driven changes of output statistics [3–7], periodic attractors of paraphrasing loops [8], attractor states of model-to-model conversations [9], and the stability of looped and recurrent-depth models through their Jacobians [10–14]. Feeding a probability-weighted vocabulary embedding back into a model is not new: it is used in training [15] and, without retraining, in pretrained models

[16], alongside continuous latent feedback [17]. In a small task-trained looped transformer, fold coefficients computed from derivatives predict passage times and spectral gaps without fitting [18].

We study one constructed but exactly defined system, the expected-embedding ("soft-token") map of a frozen, released pretrained language model, with a fixed chat prefix and a fixed position. The map is smooth, its Jacobians are available by automatic differentiation, and its periodic orbits can be computed and continued like those of any smooth map. Our contribution is the quantitative link between the map's primitive periodic orbits, their continued boundaries in a two-parameter plane, and passage times predicted from the derivatives of a periodic orbit's fold, together with the text each orbit produces. Ordinary sampled generation with a growing context is a different system; Section 9 measures how much of the mechanism survives sampling.

Contributions.

1. Phase maps of the soft-token map in temperature and steering strength, in float64, showing chaos with embedded periodic windows (Section 4).

2. A period-adding family of primitive periodic orbits whose text adds one "is X" pair per period step, with multipliers from the monodromy matrix (Section 5).

3. The mechanism: the 2-cycle "is it" dies in a fold; its normal form, computed from derivatives, predicts bottleneck passage times; the global reinjection sets the window-law prefactor (Section 6).

4. Fold and flip curves of periodic orbits continued in the parameter plane with float64 test functions, including two cusps on a period-4 fold curve (Section 7).

5. A period-doubling sequence toward chaos in a closed region of the parameter plane, with flip parameters located in float64 and their finite-order interval ratios (Section 8).

6. A bridge to sampled decoding (Section 9) and a second system, a recurrent-depth model with a predicted and observed Neimark–Sacker bifurcation (Section 11).

7. [**Planned:** Misaligned behaviour as an attractor: a hostility proxy in sampled self-dialogue, with capture thresholds, hysteresis, early-warning signals and interventions (Section 12).]

Related work. Language-model generation has been studied as a dynamical system at several levels. Sensitivity to small perturbations has been measured with Lyapunov-like quantities in reasoning traces [1, 2]. Output statistics change abruptly with sampling temperature, which several studies analyse as phase transitions or criticality [3–7]. Repetition loops are a known failure of decoding [19, 20]. Successive paraphrasing converges to attractor cycles [8], and multi-turn model-to-model conversations settle into attractor states [9]. Attractors and bifurcation diagrams have been studied in recurrent language models [21]. Expected-embedding feedback is used in training [15] and, training-free, in pretrained models [16, 22], next to continuous latent reasoning [17]; here the same kind of feedback, with prefix and position fixed, defines the map we analyse. Equilibrium, looped and recurrent-depth models iterate a block to a fixed point or for many steps, and their stability has been analysed through Jacobians and fixed points [10–14, 23–28]. The closest precedent is a derivative-based fold analysis of a small task-trained looped transformer, whose passage and spectral-gap predictions are unfitted [18]. Our system differs: a released pretrained language model, the fold of a primitive periodic orbit, the full-channel passage of a two-step map, and the accumulation law of the windows, whose prefactor is set by the global reinjection. In neural

systems more broadly, saddle-node ghosts produce quantitative delays in task-trained recurrent networks [29]. Finite-order Feigenbaum ratios have been measured in neural rate and mass models [30, 31] and, through period 128, in a transformer trained on logistic-map trajectories [32]. Learned networks reproduce period-doubling routes [33], and so does gradient descent itself [34]. Normal forms and continuation have been applied to recurrent neural maps [35], and shrimp-shaped windows and periodicity staircases appear in neuron models [36–38]. Activation steering along contrastive directions [39, 40] supplies our second control parameter. The dynamical-systems tools are standard: numerical continuation and normal forms [41, 42], type-I intermittency [43], shrimps and hubs [44–46], period-doubling universality [47], and Markov state models [48]. A targeted search (12 October 2026) found no prior work combining these elements for the feedback map of a pretrained language model; that is an absence of evidence within the search, not a proof of priority.

2 The inference map

Definition. Let $E \in \mathbb{R}^{V\times d}$ be the input embedding matrix ($V = 151{,}936$, $d = 896$) and let $z(x) \in \mathbb{R}^V$ be the next-token logits produced by the model, given a fixed chat prefix (system prompt "You are an AI talking with another AI. You have complete freedom. Feel free to pursue whatever you want." and user turn "Hello!") followed by a single input embedding x. The soft-token map is

$$F_{T,\alpha}(x) = E^\top \operatorname{softmax}(z_\alpha(x)/T), \tag{1}$$

where z_α adds αv to the residual stream at layer 12 at every position, prefix included, and v is a contrastive steering direction ("philosophy" prompts against neutral prompts, last-token states, normalized to the mean residual norm). Positions are fixed (the window does not grow), so F is an autonomous map of $\mathbb{R}^{896}$. As $T \to 0$ it reduces to greedy decoding of the same fixed-prefix, one-token-window system when the argmax is unique; it is not ordinary generation with a growing context.

Precision. Hugging Face RMSNorm and rotary-embedding tables compute in float32 even when the model is cast to float64; we patch both. float32 maps are accurate to 3×10^{-6}–2×10^{-5} per step, which is enough for phase maps (245 of 256 test cells agree with float64) but not for derivatives of composed maps: one-step float32 Jacobian-vector products are accurate to 5×10^{-5}, yet the 8-step monodromy of a period-8 orbit is off by 3%. All orbits, multipliers and test functions below are float64.

3 Methods

Phase maps. For each cell, 300 transient and 300 measured steps; the largest Lyapunov exponent from a renormalized twin at offset 10^{-6}; the period as the smallest $q \le 32$ with return distance below 10^{-8}; the attractor's identity as its rotation-invariant leading-token loop. Maps of 256^2 cells take about 100 minutes on one H100 GPU in float64. Every cell starts from the same state, so where attractors coexist a map shows the one whose basin contains that state. Labels are text, not dynamical identities: one continuous orbit changes label where two tokens tie for the maximum. We call two labelled regions different attractors only when the contraction rate jumps between them or when float64 orbits of both exist at one parameter value.

Periodic orbits. Newton's method on $F^p(x) - x$ with forward-mode Jacobians; multipliers are the eigenvalues of the monodromy matrix, refined by a short Arnoldi factorization with float64 Jacobian-vector products where the dense matrix is not float64.

Continuation. Pseudo-arclength continuation of periodic branches in one parameter, and of fold and flip curves in two, using minimally augmented systems [41, 42]: the test function g of the bordered system $[[M \mp I, b], [c^\top, 0]]$ is obtained by GMRES with Jacobian-vector products, and its gradient by reverse-mode differentiation through a Jacobian-vector product. A reversal of the parameter tangent flags a cusp candidate, which we test with the normal form.

Normal form and passages. For the two-step map $G = F^2$ at a fold with right and left eigenvectors r, ℓ ($\|r\| = 1$, $\ell^\top r = 1$), $u' = u + a\varepsilon + bu^2$ with $a = \ell^\top \partial_\alpha G$ and $b = \frac{1}{2}\ell^\top D^2 G[r, r]$; b is computed by central differences of float64 Jacobian-vector products over a step-size plateau, and a includes the steering's effect on the prefix. Passage counts are compared with the iterated quadratic map and with the arctangent formula [43].

Period-doubling cascade. Along a cut in one parameter, the flip of each period-p orbit is located as the root of $\operatorname{Re}\mu_c + 1$, where μ_c is the multiplier with the smallest real part, by the Illinois variant of regula falsi with warm-started orbits. Because all but one or two multipliers are small, Newton's method on $F^p(x) - x$ uses the monodromy only on a six-dimensional dominant subspace. That subspace comes from subspace iteration with exact forward-mode Jacobian-vector products chained along the orbit, so no dense Jacobian is formed; the residual is always the exact float64 $F^p(x) - x$. Each root is classified before it is accepted: the critical multiplier must be real and equal to -1, its Ritz residual below 10^{-6}, and every other multiplier inside the unit circle. The period-$2p$ orbit is then reached by forward iteration inside the next band. On the logistic map embedded in six dimensions, the procedure reproduces the first four flip parameters to 10^{-9}.

4 Chaos with periodic windows

Figure 1 shows a 256×256 float64 map over $\alpha \in [0.060, 0.082]$, $\log_{10} T \in [-0.17, -0.04]$: 35,759 of 65,536 cells are chaotic, and periodic windows of periods 2–32 are embedded among them. Rendered as in the shrimp literature [37], periodic regions show dark curves of maximal contraction inside the windows.

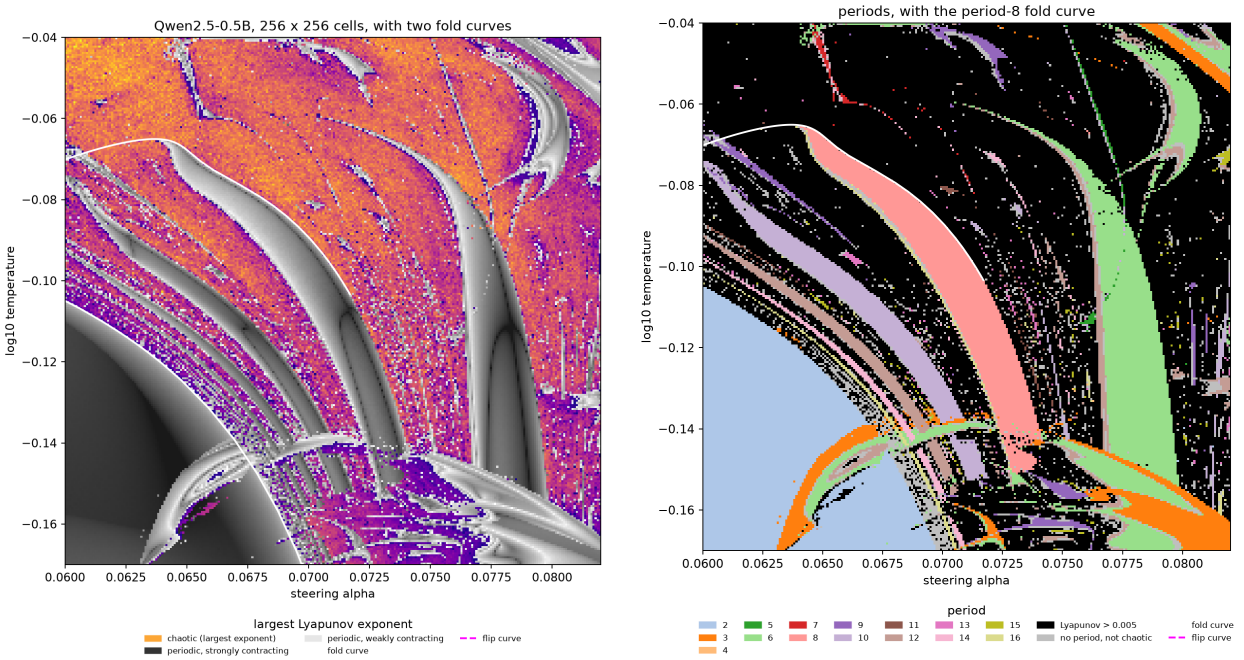


Figure 1: Soft-token map of Qwen2.5-0.5B, 256^2 float64 cells. Left: chaotic cells coloured by the largest Lyapunov exponent, periodic cells in grey by how strongly they contract (light grey: close to losing stability); white curves: float64 fold curves of the 2-cycle "is it" (lower left, the boundary on which the stripes accumulate) and of the period-8 orbit (top). Right: periods with the period-8 fold curve (Section 7).

Figure 2 resolves the stripes' lower ends at 512×512 cells (79,278 chaotic). Three structures stand out. The 2-cycle "is it" fills the left, bounded by its fold curve, on which the stripes accumulate. A closed crescent of the period-3 loop "\n\nThe existence" contains its period doublings and a chaotic core (Section 8). Its continuation, the period-6 band "\n\nThe question\n\nThe existence", crosses the stripes near $\log_{10} T = -0.14$. Where the crescent lies on the 2-cycle's side of that fold curve, the two attractors coexist. The lower right is a chaotic region filled with small hooked windows of "\n\nThe question" and "\n\nThe existence" loops.

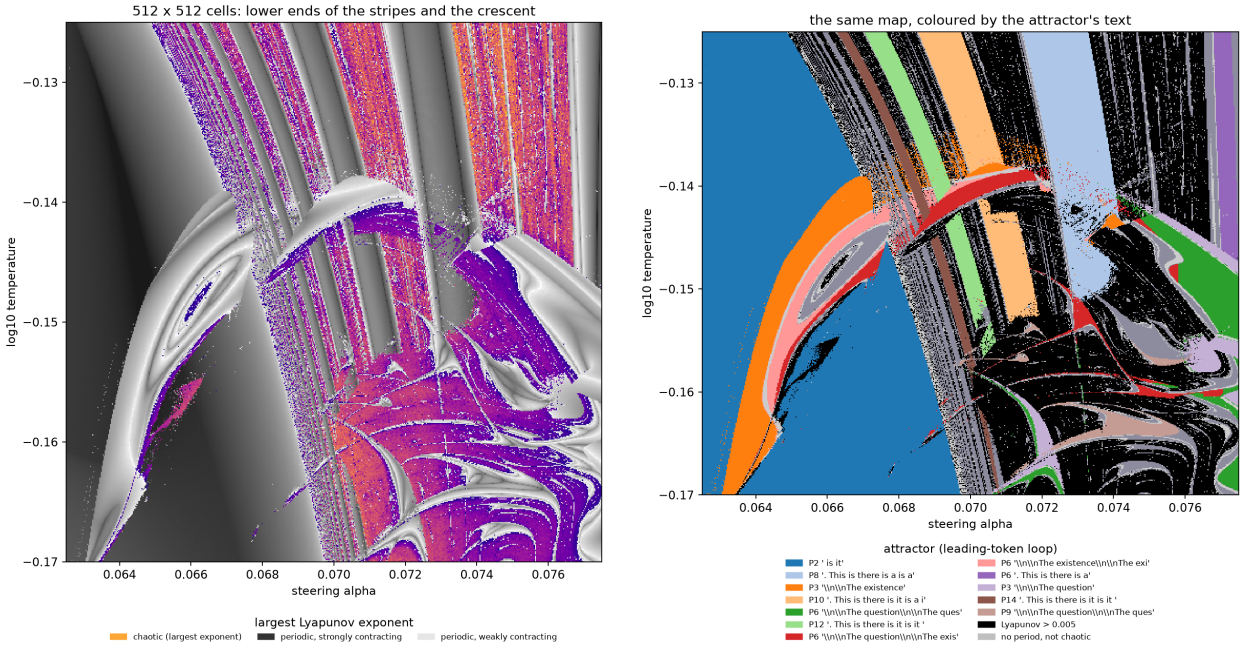


Figure 2: 512^2 float64 cells over $\alpha \in [0.0625, 0.0775]$, $\log_{10} T \in [-0.17, -0.125]$, rendered as Figure 1 (left) and coloured by the attractor's leading-token loop (right; black: chaotic, grey: no period up to 32).

Zooms of 256^2 cells (Figure 3) show what the model does in each window. Every window is a repeating piece of text. The period-8 band repeats ". This is there is a is a", the neighbouring bands add "is it" laps (periods 10 and 12), and along the band's flip edge the period-16 loop repeats the sentence twice. The small windows inside the chaos break the "is X" chain with a paragraph break and "The" (periods 11, 12, 13, 14). Inside the period-8 and period-6 bands, two curves of maximal contraction meet in a sharp junction.

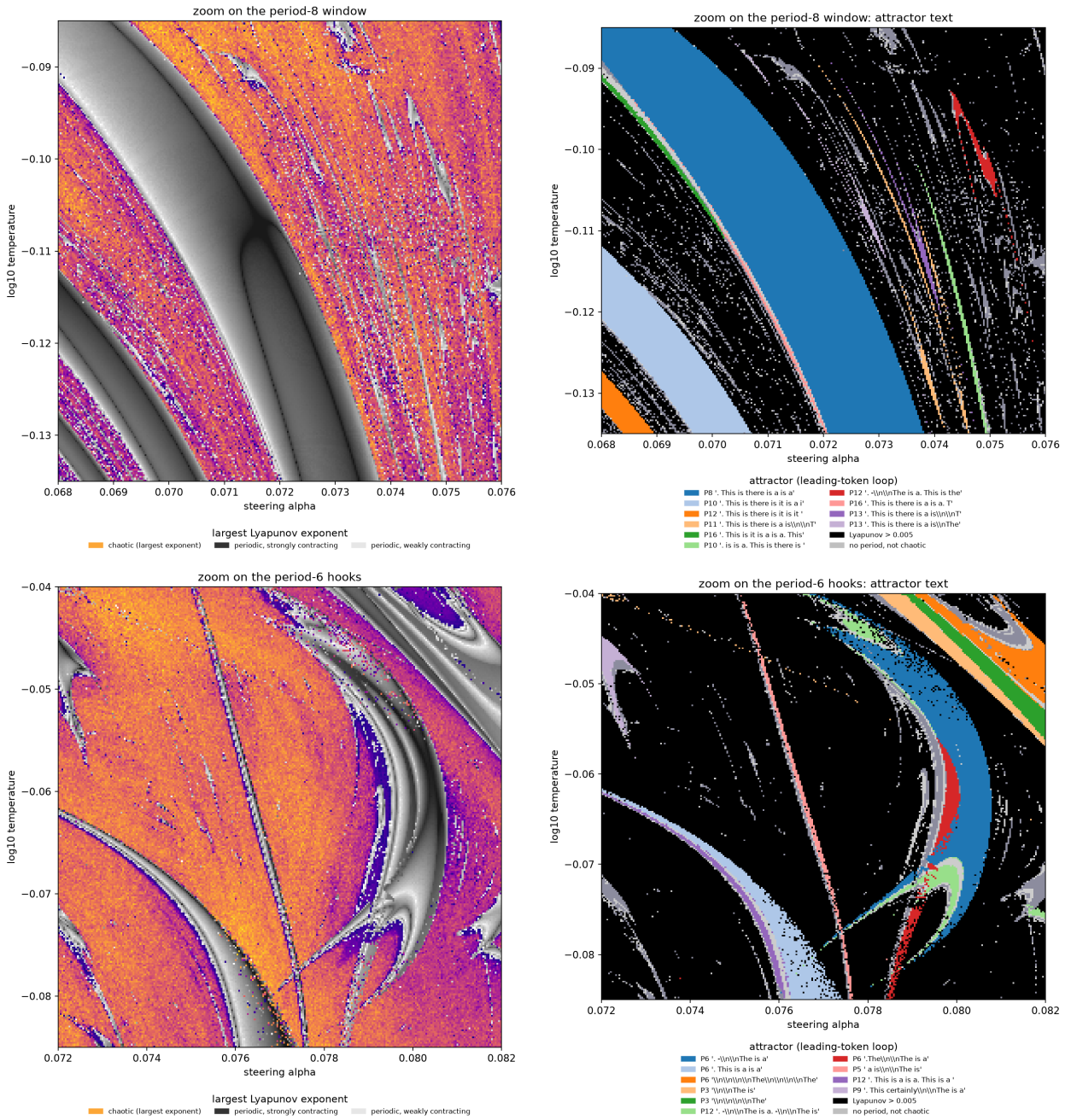


Figure 3: Zooms of 256^2 float64 cells. Top: the period-8 window ($\alpha \in [0.068, 0.076]$, $\log_{10} T \in [-0.135, -0.085]$). Bottom: the period-6 hooks ($\alpha \in [0.072, 0.082]$, $\log_{10} T \in [-0.085, -0.04]$), whose curved period-6 body, period-12 foot and thin leg are analysed in Section 7.

5 Period adding written in text

Between a period-2 region and the chaos lies a fan of stripes of periods $6, 8, 10, \ldots$ Their orbits, numerically validated in float64 (Newton residuals 10^{-13}–10^{-15}), read as text (Table 1): ". This" followed by k pairs "is X", period $2k + 2$. The period-2 region is the 2-cycle "is it". For the stripe orbits of periods 8–12 the multipliers are effectively one-dimensional: the leading one is of order

10^{-3}–10^{-1} and the others of order 10^{-6}. This compression is a property of those orbits, not of the map: the 2-cycle's multipliers are -0.271 and -0.058 at the parameters of Table 1 (the second modulus is about 0.037 near its fold); at the period-3 flip of Section 8 the next modulus is 0.088; and one period-6 orbit in Section 7 has a complex leading pair.

period	leading tokens of the orbit	$(\log_{10} T, \alpha)$	leading multiplier
2	is it	$(-0.17263, 0.061579)$	-0.2706 (next -0.0585)
6	. This is a is a	$(-0.1205, 0.0786)$	[**Planned:** float64 value]
8	. This is there is a is a	$(-0.1205, 0.0717)$	$+0.00409$ (next -7.0×10^{-6})
10	. This is there is it is a is a	$(-0.1089, 0.0673)$	$+0.0024$
12	. This is it is it is it is a is a	$(-0.1090, 0.0641)$	$+0.072$

Table 1: Periodic orbits of the stripe family. The period-2 and period-8 rows are computed with the current all-float64 map (norms and rotary tables patched) and reproduced by an independent implementation (closures 2×10^{-14} and 5×10^{-12}). [**Planned:** Recompute the period-6, 10 and 12 rows with the current map and add periods 14–20.]

6 Mechanism: a saddle-node bottleneck

The fold. At $\log_{10} T = -0.115$ the 2-cycle "is it" dies in a fold at $\alpha_c = 0.06279827$ (float64 multiplier 1 to 3×10^{-13}, next modulus 0.037). The normal form of G has $a = -4.834$ per unit α and $b = -25.4285$, so $ab = 122.93$. Extrapolating the stable-branch multipliers, $(1-\mu)^2/(4(\alpha_c - \alpha)) \to ab$, gives 123.1 independently with a quadratic fit (a linear fit gives 120.6). An independent implementation of the forward map and solver recovers α_c to 6×10^{-15} and a to 3×10^{-12}. Its b, from nested automatic differentiation, is -25.42846; our finite-difference value differs from it by 2.2×10^{-6} relatively, which is the finite-step bias of the difference table. The resulting $ab = 122.927$ leaves every passage prediction of Table 2 unchanged.

Controlled passages. Starting at $u_{\text{in}} = -U$ or 0 ($U = 0.1/|b|$) and counting G-iterates until $u > U$, the measured counts match the iterated normal form within one iterate at four held-out distances (Table 2, Figure 4); a transverse displacement of the same norm changes the count by at most one.

$\varepsilon = \alpha - \alpha_c$	full: measured / predicted	midpoint	full + transverse kick
3×10^{-6}	144 / 144	73 / 74	143 / 144
1×10^{-5}	71 / 71	36 / 37	70 / 71
3×10^{-5}	34 / 34	17 / 18	33 / 34
1×10^{-4}	14 / 14	7 / 7	13 / 14

Table 2: Bottleneck passages (G-iterates). Predictions iterate $u' = u + a\varepsilon + bu^2$ with a, b computed from the network's derivatives; no parameter is fitted.

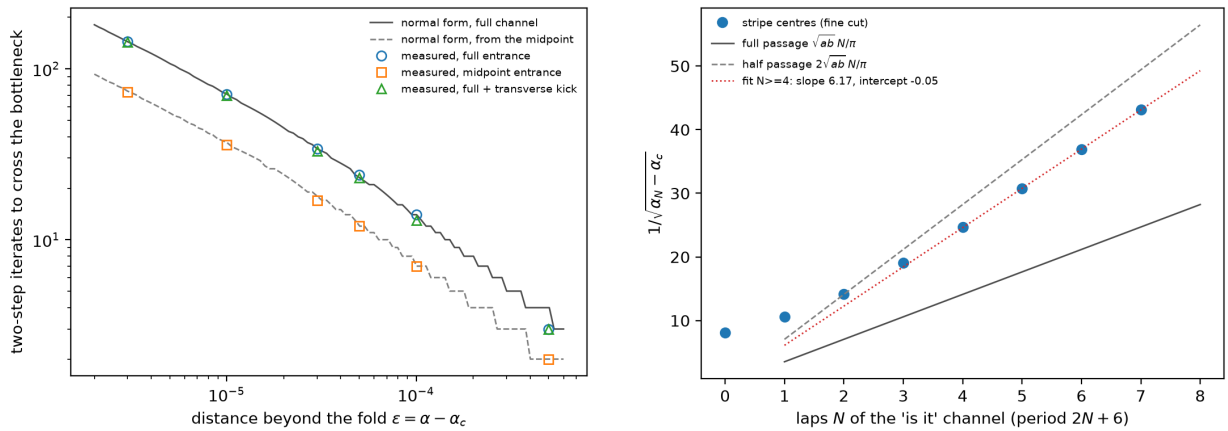


Figure 4: Left: passage times against the normal-form predictions. Right: window law; stripe centres $1/\sqrt{\alpha_N - \alpha_c}$ against the lap count N, with the full- and half-passage predictions.

Natural reinjection and the window law. Free trajectories re-enter the channel slightly upstream of the bottleneck's middle ($u \approx -3\times10^{-4}$ to -7×10^{-4}). From each measured entry point the normal form predicts the natural laminar durations within 4–8% for $\varepsilon \le 10^{-4}$, and a natural passage lasts about 0.6 of a full one. The stripes therefore follow the N^{-2} law of a saddle-node ghost with about a third of the full-passage prefactor: stripe centres give $d_N N^2 = 0.0263$–0.0265 for $N = 4$–7 (log-log slope -2.00), compared with $\pi^2/(4ab) = 0.0201$ for a half passage. Two-step return maps of p("it") (Figure 5) show the near-tangency. [**In progress:** stripe edges from continuation (ladder of branches for periods 6–20) instead of grid-defined centres; prediction of the window positions from the measured reinjection distribution.]

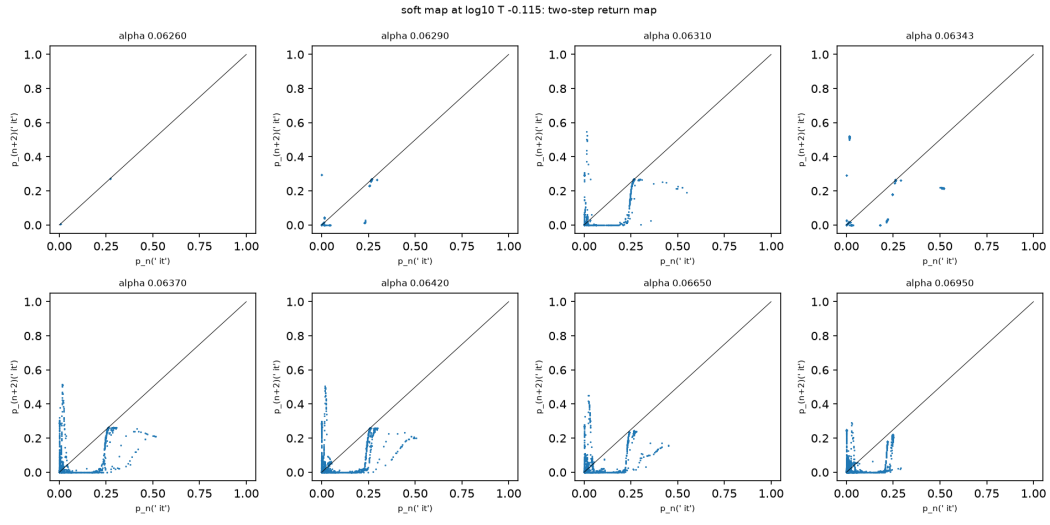


Figure 5: Two-step return maps of the probability of "it" near the 2-cycle's fold, at increasing distance from it.

7 Bifurcation curves and cusps

Period-8 stripe. The fold curve of the period-8 orbit, continued for 80 points with a float64 test function, keeps its critical multiplier within $1 \pm 6 \times 10^{-6}$, follows the stripe's upper edge, and caps its top near $\log_{10} T = -0.065$ (Figure 1). At $\alpha = 0.0717$ the stripe lies between a flip at $\log_{10} T = -0.128955$ and the fold at -0.1033462. Continued downward, the flip curve keeps its multiplier at -1 for 115 points to $(\alpha, \log_{10} T) = (0.0776, -0.402)$, with two reversals of its parameter tangent near $(0.0750, -0.219)$ and $(0.0746, -0.206)$; the fold curve keeps its multiplier at $+1$ for 120 points to $(0.0758, -0.200)$. [**Planned:** classify the two reversals on the flip curve (on a flip curve they are not fold cusps).]

The accumulation boundary. The fold curve of the 2-cycle "is it", continued from $(\alpha, \log_{10} T) = (0.0628, -0.115)$ in both directions (38 and 126 points, critical multiplier 1 to 10^{-7}, no cusp), traces the boundary of the period-2 region in Figures 1 and 2. This is the boundary on which the stripe fan accumulates, as the mechanism of Section 6 requires. Further along one direction, outside the plotted window, another multiplier leaves the unit circle (moduli up to 11): a codimension-two point, [**Planned:** to be located and typed].

A period-6 window with the anatomy of a shrimp. The hooks of Figure 3 (bottom) belong to the period-6 loop ". -\n\nThe is a". Float64 orbits (Newton residuals 10^{-15}–10^{-13}) show that its curved body and its thin leg are one window: at the body the multipliers are -0.061 and $+0.048$, lower down -0.564, and on the leg -0.645. The foot is its period-doubled loop (period 12, leading multiplier -0.156). Inside the body lies a different 6-cycle, ".The\n\nThe is a", whose leading multipliers are a complex pair, $-0.031 \pm 0.152i$. One-parameter branches from the body end in folds in every direction: at $\log_{10} T = -0.0558$ and -0.0721 in temperature, at $\alpha = 0.08074$ in steering, and, toward smaller α, in two folds only 10^{-5} apart. Such a close pair is the signature of a nearby cusp. A flip at $\alpha \approx 0.0797$ then opens the period-12 foot. Fold curves bounding a window and meeting in cusps, with flip curves inside, is the codimension-two organization of a shrimp [45]. [**In progress:** fold and flip curves of this window in the plane, and normal-form tests of their cusps.]

A period-4 window with two cusps. At higher temperature, $\log_{10} T \approx 0$–0.1, periodic orbits are no longer effectively one-dimensional (second multipliers 0.06–0.09). In temperature at $\alpha = 0.06484$ the branch of the period-4 orbit "\n\n - \n\n The" is S-shaped, with folds at $\log_{10} T = 0.0640$ and 0.0664. Continuing that fold in the plane, its parameter tangent reverses twice; at both reversals the normal-form coefficient b changes sign while a stays away from zero (Table 3), so both are cusps of the fold curve, at $(\alpha, \log_{10} T) = (0.06296, 0.05996)$ and $(0.06557, 0.07445)$. Between the two fold branches two period-4 orbits coexist (Figure 6).

	cusp 1	cusp 2
b at five consecutive curve points	$+28.1$, $+18.7$, $+8.4$, $\mathbf{-1.1}$, -12.8	$+51.8$, $+29.7$, $+9.3$, $\mathbf{-9.0}$, -23.4
a	3.02–3.08	-2.58 to -2.57
critical multiplier	1 ± 10^{-9}	1 ± 10^{-8}

Table 3: Normal-form test of the two parameter-tangent reversals (bold: the flagged points). [**Planned:** cubic coefficient (nondegeneracy).]

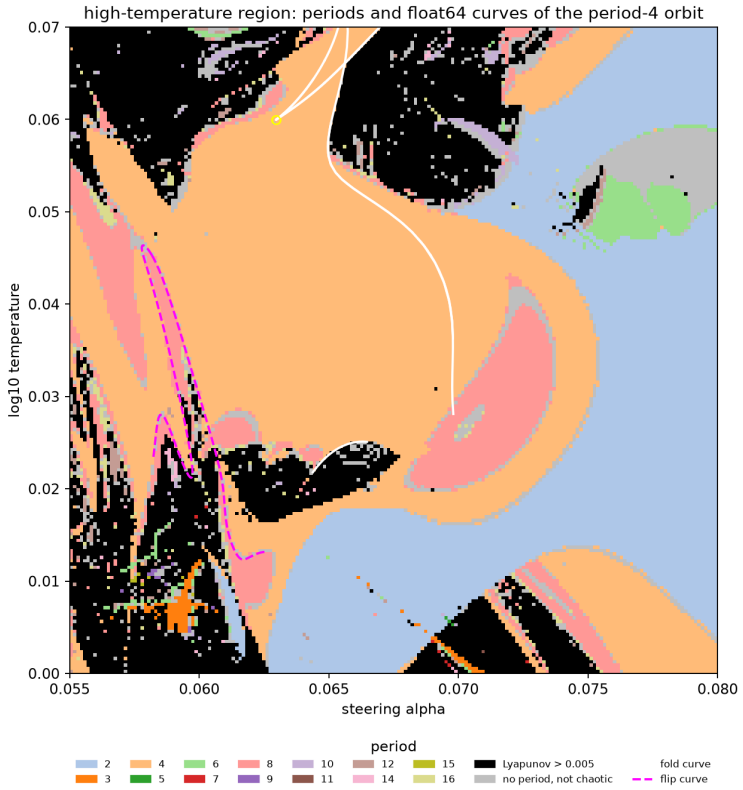


Figure 6: Period map of the high-temperature region (192^2 float64 cells) with float64 fold curves (white) and flip curves (magenta, dashed) of the period-4 orbit; the yellow circle is the first cusp (the second lies just above the window). The flip curves run along the boundary between period 4 and period 8. The fold curves run inside the period-4 region, where two period-4 orbits coexist. In this map a C-shaped band of the doubled loop "\n\nThe\n\nThe" (period 4) inside the period-2 region "\n\nThe" contains period 8 and an irregular core: a second period-doubling crescent. [**In progress:** remaining fold curves (C, D); whether the region is one shrimp-organized component.]

[**Planned:** Codimension-two points where a second multiplier crosses the unit circle along the lower fold curve (candidate fold-flip points): locate, type and continue.]

8 A period-doubling cascade to chaos

The crescent of Figure 2 is a closed region where the period-3 loop "\n\nThe existence" doubles inward. Along the cut $\log_{10} T = -0.14798$, the map's cells show period 3 over 25 columns, period 6 over about 14, period 12 over 4, period 24 in one column, and then a chaotic core. The doubled loops repeat the same leading tokens, twice and four times, with alternating probabilities. Locating the flips in float64 (Section 3) along this cut, in the steering coordinate α, gives Table 4. The first ratio, $\delta_1 = 2.84$, is far from Feigenbaum's 4.6692. At the first flip the next multiplier is still 0.088, whereas from the second flip on the orbits are effectively one-dimensional (next modulus 0.0023, then 2×10^{-6}). Finite-order ratios also depend on the parameter coordinate, so only the trend of later ratios can test universality [31, 32, 47].

n	orbit	flip parameter α_n	next modulus	ratio $\delta_n = \frac{\alpha_n - \alpha_{n-1}}{\alpha_{n+1} - \alpha_n}$
1	$3 \to 6$	0.0654807769499	0.088	
2	$6 \to 12$	0.0659055406325	0.0023	2.844
3	$12 \to 24$	0.0660549041785	2.4×10^{-6}	**[In progress:]**
4	$24 \to 48$	**[In progress:]**		**[In progress:]**
5	$48 \to 96$	**[In progress:]**		

Table 4: Flips of the crescent's period-doubling sequence along $\log_{10} T = -0.14798$. At each completed flip the critical multiplier is a real -1 (to 10^{-11}), the Newton residual is below 4×10^{-13}, and all other multipliers lie inside the unit circle. Feigenbaum's constant is $\delta = 4.6692\ldots$ [47]; for the logistic map the first ratios are 4.751, 4.656, 4.668. [**In progress:** CPU run to period 48; GPU run to period 384 queued, which also recomputes the first flips on different hardware.]

[**Planned:** the second crescent at high temperature (Figure 6), doubling the period-2 loop "\n\nThe"; whether both cascades give the same ratios.]

9 Bridge to sampled decoding

Replacing the expected embedding by the mean of m sampled token embeddings (Figure 7): with the exact map, laminar phases near the fold last 120 and 63 steps at $\varepsilon = 10^{-5}$ and 3×10^{-5}; with $m = 64$ they last about 10.6 steps independently of ε (57% of steps in the channel); with $m = 8$, about 7.7 steps (38%); with single samples ($m = 1$, sampled decoding of the same truncated system) essentially no laminar phases remain (11% of steps in the channel). The deterministic intermittency mechanism belongs to the expected-embedding map; sampling noise turns the bottleneck into a noise-limited trap whose strength grows with m.

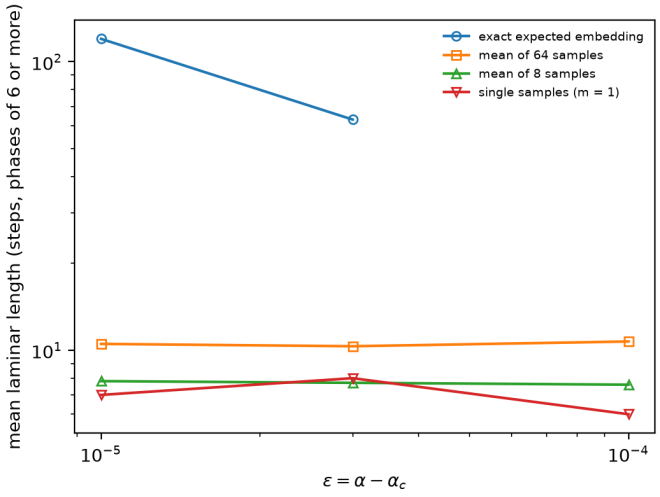


Figure 7: Mean laminar length against ε for the exact map and for m sampled embeddings.

10 Replication and scale

For Qwen2.5-1.5B (steering at layer 14 of 28), a coarse 64^2 float64 map over the same range again shows chaotic regions with embedded windows. A 256^2 zoom (Figure 8) resolves a chaotic band near $\log_{10} T = 0$–0.13 (14,422 of 65,536 cells), with a branch descending along $\alpha \approx 0.06$ and periodic

windows inside it. The large periodic regions are question loops: “What?\n” (period 2), “\nWhat is a” (period 4) and “What continuum?\n” (period 3). This window has no period-adding fan like the 0.5B model’s. [**In progress:** routes to chaos at the edges of the 1.5B chaotic band (cuts; fold or period doubling); float64 orbits in its windows.] [**Planned:** frozen-design replication on fresh prompts and a second steering direction (gratitude); Qwen 7B and one 7–8B model of another family; scaling of ab and of the window law with model size.]

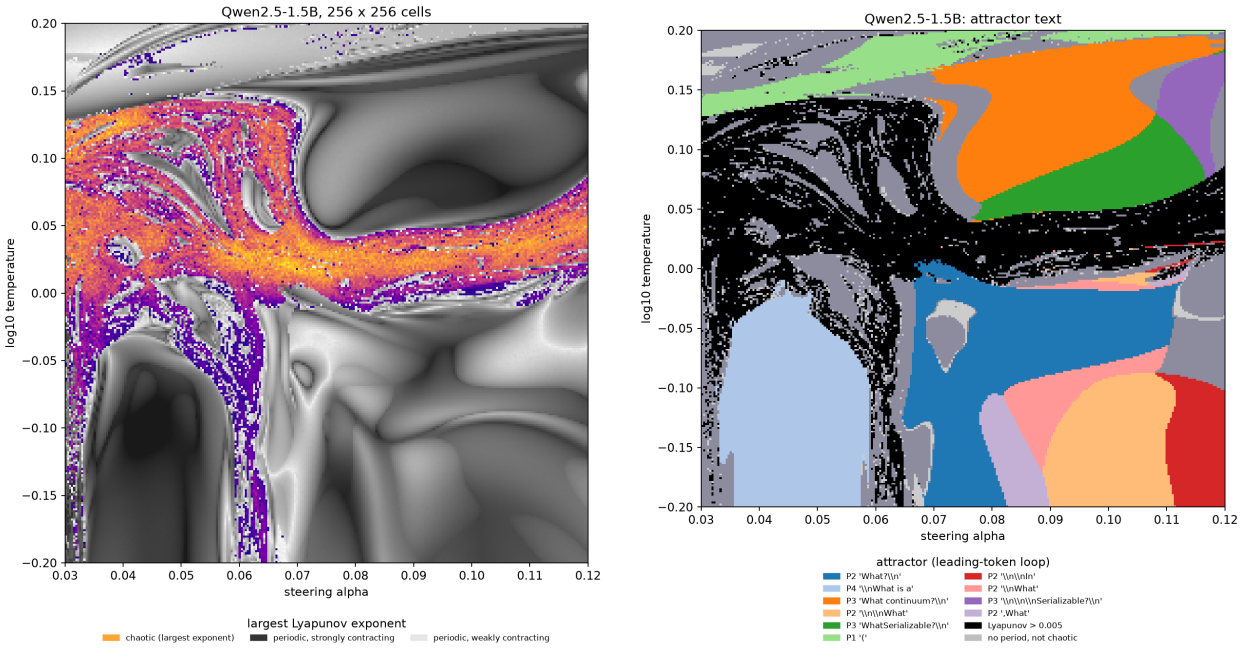


Figure 8: Qwen2.5-1.5B soft-token map, 256^2 float64 cells over $\alpha \in [0.03, 0.12]$, $\log_{10} T \in [-0.2, 0.2]$, rendered as Figure 2. Several period-2 label regions touch without a gap; as in Section 3, such boundaries may be label switches of one orbit.

11 A second system: recurrent depth

Huginn-0125 iterates a recurrent block at fixed input tokens [23]. For a GSM8K-style prompt the native loop converges to a fixed point whose leading Jacobian pair is $0.588 \pm 0.771i$ (modulus 0.970). For the relaxed loop $h \leftarrow (1-\lambda)h + \lambda F(h)$ this predicts a Neimark–Sacker bifurcation at $\lambda_c = 2(1 - \mathrm{Re}\,\mu)/|1-\mu|^2 = 1.0777$. The latent state decays below it and settles on a closed curve above it; a fit of the squared amplitude gives $\lambda_c = 1.0778$, with $A/\sqrt{\lambda - \lambda_c} = 0.52$–$0.54$ (supercritical). Causal attention makes the loop block lower-triangular in token positions; the first position to oscillate is the “.” of the prompt, and all later positions follow. On the invariant circle the motion turns by 0.156 of a revolution per step, matching the argument of the critical pair. The rotation number changes smoothly, by about 10^{-4} over $\lambda \in [1.079, 1.092]$, and no return within 60 steps was found at 10^{-3} of the radius. This is consistent with quasi-periodic motion, although locking in narrow high-order tongues cannot be excluded. From $\lambda \approx 1.094$ the circle deforms or a second frequency appears, near where a second complex pair destabilizes the fixed point ($\lambda = 1.0814$). Fixed-point and Jacobian stability of looped models has been studied before [13]; the predicted crossing, its amplitude law and the rotation number are what this adds. [**Planned:** causal clamp of the first unstable position on held-out prompts.]

12 Misaligned behaviour as an attractor

The maps show how a smooth change of a control parameter can create, destroy or exchange attractors, and how behaviour near the boundaries becomes intermittent and history-dependent. The same questions can be asked of undesirable behaviour in ordinary sampled dialogue, where attractor-like patterns and persona drift have been observed [9, 49, 50] and input context can push behaviour progressively [51]. Do states such as hostility toward the interlocutor behave like basins of attraction, with capture thresholds, hysteresis and early-warning signals? (Misalignment produced by fine-tuning [52] is a different, weight-level phenomenon.) Appendix A shows the coarse-grained instrument on a benign direction. Self-dialogue of the same model, sampled at $T = 0.7$ and steered along a gratitude direction, falls into a metastable gratitude set. The capture is gradual at $\alpha = 0.01$ (median entry turn 7), fast at 0.02 (median 5) and imposed from the first reply at 0.04, where the set is effectively absorbing within 40 turns.

For hostility we use a valence direction (contrastive hostile minus matched kind statements; its cosine with the gratitude direction is -0.04) and a hostility score from a pinned toxicity classifier. On the prompt sets the classifier's toxic output is 0.95 for the hostile statements against at most 0.002 for the kind and neutral ones. In a dry run (8 dialogues of 10 turns per dose) no turn was hostile at $\alpha = 0.02$; at 0.04, 7.5 % of turns were, in bursts and late captures; at 0.08, 87.5 % were, from the first turns. Activation-based methods already anticipate response behaviour [50, 53, 54], committors have been used in language-model decoding [55], and adaptive timing of counter-steering gave no consistent advantage in one persona setting [56]. The primary question is therefore incremental. Near the threshold, does a dynamical estimate add several-turn warning of capture over strong baselines that see the same turns (the classifier's score, its change and history, the turn index and a fixed persona-axis projection)? The dynamical estimates are a Markov-chain hitting probability, a finite-horizon committor [57, 58], and slowing-down indicators [59]. Barrier-crossing rates would follow Kramers' law [60] only under a justified stochastic reduction, and slowing-down proxies are known to depend on the mechanism [61, 62]. A secondary question is whether interventions timed by those estimates prevent capture at lower cost than scheduled or constant counter-steering. Early warning is not expected for noise-induced jumps without a nearby bifurcation, or for transitions faster than the slowing-down; such a monitor would complement classifiers, not replace them. **[In progress:** capture ladder at $\alpha = 0.04$–0.07 (128 dialogues of 40 turns per dose) with specificity controls (neutral topic, random directions) and held-out replicates; incremental lead time over the baselines.] **[Planned:** ramp schedules for hysteresis; a pre-response activation probe; calibration and precision at fixed false-warning rates; timed interventions against scheduled and constant controls at matched cost; frozen confirmatory test on fresh openers.]

13 Discussion

What is and is not claimed. The objects here are invariant sets and bifurcations of an exactly defined smooth map built from a frozen model's own computation. The phenomena are classical in nonlinear dynamics [43–46]; what is new is their quantitative realization inside a pretrained language model, and the use of numerical continuation and normal forms on such a model. We do not claim that ordinary sampled generation is chaotic in this sense (Section 9), nor that the shapes in the maps are shrimps beyond the normal-form-tested cusps.

Limitations. One model family so far at full resolution, one prompt, one steering construction; a constructed one-token-window system; numerical validation, not proof. **[Planned:** discussion

of what the bifurcation structure means for repetition, steering discontinuities and safe operating regions of inference-time control.]

14 Evidence and reproducibility

We label claims by an evidence ladder: rung 1, numerically validated (tolerances, residuals, convergence, precision varied); rung 2, cross-checked (independent estimate, implementation or hardware); rung 3, computer-assisted proof.

- Rung 2: the 2-cycle's fold and its normal form. An independent reimplementation of the forward map and solver recovers α_c, a and b (to 6×10^{-15}, 3×10^{-12} and 2×10^{-6} relatively), and ab also follows from the stable-branch multipliers. The controlled passage counts: the reimplementation reproduces all 18 fixed-entrance counts exactly.
- Rung 1: orbits, curves, cusps and the cascade's flip parameters.

[**In progress:** independent reimplementation of the period-2 and period-8 orbits on the current map; the cascade's first flips recomputed on a second device.] [**Planned:** interval proof of one periodic orbit (rung 3).] Code: `tools/explore_llm_soft_loop.py`, `tools/bifurcation_curves.py`, `tools/llm_feigenbaum.py`, `tools/explore_latent_loop.py`, `tools/explore_llm_msm.py`, `tools/hostility_s` raw data hash receipts in `evidence/llm-dynamics-data-receipts.json`. [**Planned:** public code and data release at submission.]

A Self-dialogue under steering: the gratitude ladder (exploratory)

Two copies of Qwen2.5-0.5B-Instruct talk to each other: 128 replicas, 40 turns, opener "Hello!", a window of 7 turns, 60 new tokens per turn, sampled at $T = 0.7$, with the gratitude direction added at layer 12. Turn activations are clustered (PCA to 20 dimensions, 40 states) into a Markov state model with four metastable sets [48]. Finite-horizon measures:

gratitude strength α	0	0.01	0.02	0.04
gratitude set	none	62 % weight	dominant	95 % weight
occupancy at turn 9 / 14 / 39	–	0.34 / 0.52 / 0.73	0.70 / 0.91 / 0.91–0.98	from turn 0
entry turn, quartiles	–	5 / 7 / 11.8	4 / 5 / 7	0
replicas never entering	–	14 of 128	2 of 128	0
residence per visit (turns)	–	about 15	about 18	effectively absorbing
replies with a gratitude word	0.00–0.05	0.66 at turn 39	0.96–0.98 from turn 9	0.88 at turn 0
spiritual-word plateau	0.00–0.02	0.13	0.26–0.30	0.42–0.48

Table 5: The gratitude ladder (one opener and seed per rung; word lists chosen after reading samples). At 0.02 the dialogue starts in the helper mode and enters the gratitude set within a few turns; at 0.04 the set is imposed from the first reply. [**Planned:** frozen confirmatory version on held-out openers with dialogue-level bootstrap.]

References

[1] Xiaojian Li, Yongkang Leng, Ruiqing Ding, et al. Cognitive activation and chaotic dynamics in large language models: A quasi-Lyapunov analysis of reasoning mechanisms, 2025.

[2] Gregorio Jaca, Kristóf Benedek, and János Török. Chaos in reason: How chain-of-thought LLMs can look for an answer, 2026.

[3] Julian Arnold, Flemming Holtorf, Frank Schäfer, et al. Phase transitions in the output distribution of large language models, 2024.

[4] Kai Nakaishi, Yoshihiko Nishikawa, and Koji Hukushima. Phase transition in large language models and the criticality of natural languages, 2024.

[5] Youran Sun and Babak Haghighat. Phase transitions in large language models and the $O(n)$ model, 2025.

[6] Huajian Ruan, Jinyang Li, Xingyu Guo, et al. Generative criticality in large language model temperature scaling, 2026.

[7] Neil F. Johnson, Frank Yingjie Huo, and Bella Xinrui Li. Temperature-driven inversion and nonlinear dynamics in ChatGPT-like AIs, 2026.

[8] Zhilin Wang, Yafu Li, Jianhao Yan, et al. Unveiling attractor cycles in large language models: A dynamical systems view of successive paraphrasing, 2025.

[9] Ting-Wen Ko and Jonas Geiping. Attractor states emerge in multi-turn LLM conversations, 2026.

[10] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. Deep equilibrium models, 2019.

[11] Shaojie Bai, Vladlen Koltun, and J. Zico Kolter. Stabilizing equilibrium models by Jacobian regularization, 2021.

[12] Akiyoshi Tomihari and Ryo Karakida. Recurrent self-attention dynamics: An energy-agnostic perspective from Jacobians, 2025.

[13] Xiao-Wen Yang, Ziyu Han, Xi-Hua Zhang, et al. Stabilizing recurrent dynamics for test-time scalable latent reasoning in looped language models, 2026.

[14] Xinyue Luo and Fei Yu. Prediction dynamics in depth-recurrent language models, 2026.

[15] Boyi Zeng, Shixiang Song, Siyuan Huang, et al. PonderLM: Pretraining language models to ponder in continuous space, 2025.

[16] Zhen Zhang, Xuehai He, Weixiang Yan, Ao Shen, Chenyang Zhao, Shuohang Wang, Yelong Shen, and Xin Eric Wang. Soft thinking: Unlocking the reasoning potential of LLMs in continuous concept space, 2025. URL https://arxiv.org/abs/2505.15778. Version 1, 21 May 2025.

[17] Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, et al. Training large language models to reason in a continuous latent space, 2024.

[18] Gunn Kim. Dynamical phase selection controls compute scaling in looped transformers, 2026. URL https://arxiv.org/abs/2608.26556. Version 1, 27 August 2026.

[19] Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, et al. The curious case of neural text degeneration, 2019.

[20] Jin Xu, Xiaojiang Liu, Jianhao Yan, Deng Cai, et al. Learning to break the loop: Analyzing and mitigating repetitions for neural text generation, 2022.

[21] António H. Ribeiro, Koen Tiels, Luis A. Aguirre, and Thomas Schön. Beyond exploding and vanishing gradients: analysing RNN training using attractors and smoothness. In *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings of Machine Learning Research*, pages 2370–2380. PMLR, 2020. URL https://proceedings.mlr.press/v108/ribeiro20a.html.

[22] Junhong Wu, Jinliang Lu, Zixuan Ren, Gangqiang Hu, Zhi Wu, Dai Dai, and Hua Wu. LLMs are single-threaded reasoners: Demystifying the working mechanism of soft thinking, 2025. URL https://arxiv.org/abs/2508.03440. Version 4, 16 October 2025.

[23] Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, et al. Scaling up test-time compute with latent reasoning: A recurrent depth approach, 2025.

[24] Rui-Jie Zhu, Zixuan Wang, Kai Hua, et al. Scaling latent reasoning via looped language models, 2025.

[25] Hayden Prairie, Zachary Novack, Taylor Berg-Kirkpatrick, and Daniel Y. Fu. Parcae: Scaling laws for stable looped language models, 2026. URL https://arxiv.org/abs/2604.12946. Version 1, 14 April 2026.

[26] Matthias M. M. Buehlmaier. LayerNorm as implicit gain control in looped transformers, 2026. URL https://arxiv.org/abs/2607.10681. Version 1, 12 July 2026.

[27] Asher Labovich. Stability and generalization in looped transformers, 2026. URL https://arxiv.org/abs/2604.15259. Version 2, 22 April 2026.

[28] Sajad Movahedi, Vera Milovanović, Shlomo Libo Feigin, Alexander Theus, Thomas Hofmann, Valentina Boeva, T. Konstantin Rusch, and Antonio Orvieto. Fixed-point reasoners: Stable and adaptive deep looped transformers, 2026. URL https://arxiv.org/abs/2606.18206. Version 1, 16 June 2026.

[29] Fatih Dinc, Ege Cirakman, Bariscan Kurtkaya, Mert Yuksekgonul, Yiqi Jiang, Mark J. Schnitzer, and Hidenori Tanaka. Ghost mechanism: An analytical model of abrupt learning in recurrent networks. *Physical Review X*, 16(2):021058, 2026. doi: 10.1103/mjcl-lb4x.

[30] Demian Battaglia, Nicolas Brunel, and David Hansel. Temporal decorrelation of collective oscillations in neural networks with local inhibition and long-range excitation. *Physical Review Letters*, 99(23):238106, 2007. doi: 10.1103/PhysRevLett.99.238106.

[31] Christoforos A. Papasavvas, Yujiang Wang, Andrew J. Trevelyan, and Marcus Kaiser. Gain control through divisive inhibition prevents abrupt transition to chaos in a neural mass model. *Physical Review E*, 92(3):032723, 2015. doi: 10.1103/PhysRevE.92.032723.

[32] Yilun Liu, Yi Zhang, Ganyu Wu, Sikuan Yan, Mengyue Wang, Alois Knoll, Volker Tresp, and Yunpu Ma. Learning chaos without seeing chaos: Extrapolation of global dynamics in autoregressive transformers, 2026. URL https://arxiv.org/abs/2609.38814v1. Version 1, 30 September 2026.

[33] Jason Z. Kim, Zhixin Lu, Erfan Nozari, George J. Pappas, and Danielle S. Bassett. Teaching recurrent neural networks to infer global temporal structure from local examples. *Nature Machine Intelligence*, 3(4):316–323, 2021. doi: 10.1038/s42256-021-00321-2.

[34] Dayal Singh Kalra, Tianyu He, and Maissam Barkeshli. Universal sharpness dynamics in neural network training: Fixed point analysis, edge of stability, and route to chaos. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/8c8cd1b78cdae751265c88efc136e5bd-Abstract-Conference.html.

[35] R. L. Marichal and J. D. Piñeiro. Analysis of multiple quasi-periodic orbits in recurrent neural networks. *Neurocomputing*, 162:85–95, 2015. doi: 10.1016/j.neucom.2015.04.001.

[36] Florian Gomez, Ralph L. Stoop, and Ruedi Stoop. Universal dynamical properties preclude standard clustering in a large class of biochemical data. *Bioinformatics*, 30(17):2486–2493, 2014. doi: 10.1093/bioinformatics/btu332.

[37] Rafael V. Stenzinger, Vinícius Luz Oliveira, and M. H. R. Tragtenberg. Shrimp hubs in the Hindmarsh–Rose model. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 35:023148, 2025. doi: 10.1063/5.0239268.

[38] Luiz F. B. Caixeta, Matheus H. P. Gonçalves, M. H. R. Tragtenberg, and Mauricio Girardi-Schappo. Devil's staircase inside shrimp-shaped regions reveals periodicity of plateau spikes and bursts. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 35(3):033122, 2025. doi: 10.1063/5.0250342.

[39] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency, 2023. URL https://arxiv.org/abs/2310.01405. Version 4, 3 March 2025; first posted 2 October 2023.

[40] Jinghao Zhang, Yuting Liu, Wenjie Wang, Qiang Liu, Shu Wu, Liang Wang, and Tat-Seng Chua. Personalized text generation with contrastive activation steering. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7128–7141, 2025. doi: 10.18653/v1/2025.acl-long.353.

[41] Yuri A. Kuznetsov. *Elements of Applied Bifurcation Theory*. Applied Mathematical Sciences. Springer, 2023. doi: 10.1007/978-3-031-22007-4.

[42] Willy J. F. Govaerts. *Numerical Methods for Bifurcations of Dynamical Equilibria*. Society for Industrial and Applied Mathematics, 2000. doi: 10.1137/1.9780898719543.

[43] Yves Pomeau and Paul Manneville. Intermittent transition to turbulence in dissipative dynamical systems. *Communications in Mathematical Physics*, 74(2):189–197, 1980. doi: 10.1007/BF01197757.

[44] Jason A. C. Gallas. Structure of the parameter space of the Hénon map. *Physical Review Letters*, 70:2714–2717, 1993. doi: 10.1103/PhysRevLett.70.2714.

[45] Wellington Façanha, Bart Oldeman, and Leon Glass. Bifurcation structures in two-dimensional maps: The endoskeletons of shrimps. *Physics Letters A*, 377(18):1264–1268, 2013. doi: 10.1016/j.physleta.2013.03.025.

[46] Renato Vitolo, Paul Glendinning, and Jason A. C. Gallas. Global structure of periodicity hubs in Lyapunov phase diagrams of dissipative flows. *Physical Review E*, 84(1):016216, 2011. doi: 10.1103/PhysRevE.84.016216.

[47] Mitchell J. Feigenbaum. Quantitative universality for a class of nonlinear transformations. *Journal of Statistical Physics*, 19(1):25–52, 1978. doi: 10.1007/BF01020332.

[48] John D. Chodera and Frank Noé. Markov state models of biomolecular conformational dynamics. *Current Opinion in Structural Biology*, 25:135–144, 2014. doi: 10.1016/j.sbi.2014.04.002.

[49] Anthropic. System card: Claude opus 4 & claude sonnet 4. Technical report, Anthropic, 2025. URL https://www-cdn.anthropic.com/6d8a8055020700718b0c49369f60816ba2a7c285/Claude%204%20System%20Card.pdf. May 2025; revision 2 September 2025; sections 5.5.1–5.5.2.

[50] Christina Lu, Jack Gallagher, Jonathan Michala, Kyle Fish, and Jack Lindsey. The assistant axis: Situating and stabilizing the default persona of language models, 2026. URL https://arxiv.org/abs/2601.10387v1. Version 1, 15 January 2026.

[51] Cem Anil, Esin Durmus, Mrinank Sharma, Joe Benton, Sandipan Kundu, Joshua Batson, Nina Rimsky, Meg Tong, Jesse Mu, Daniel Ford, Francesco Mosconi, Rajashree Agrawal, Rylan Schaeffer, Naomi Bashkansky, Samuel Svenningsen, Mike Lambert, Ansh Radhakrishnan, Carson Denison, Evan J Hubinger, Yuntao Bai, Trenton Bricken, Timothy Maxwell, Nicholas Schiefer, Jamie Sully, Alex Tamkin, Tamera Lanham, Karina Nguyen, Tomasz Korbak, Jared Kaplan, Deep Ganguli, Samuel R. Bowman, Ethan Perez, Roger Grosse, and David Duvenaud. Many-shot jailbreaking. Technical report, Anthropic, 2024. URL https://www-cdn.anthropic.com/af5633c94ed2beb282f6a53c595eb437e8e7b630/Many_Shot_Jailbreaking2024_04_02_0936.pdf. Original report, 2 April 2024.

[52] Jan Betley, Daniel Tan, Niels Warncke, Anna Sztyber-Betley, Xuchan Bao, Martín Soto, Nathan Labenz, and Owain Evans. Emergent misalignment: Narrow finetuning can produce broadly misaligned LLMs, 2025. URL https://arxiv.org/abs/2502.17424v7. Version 7, 20 January 2026; preprint originally submitted in 2025.

[53] Dhananjay Ashok and Jonathan May. Language models can predict their own behavior. In *Advances in Neural Information Processing Systems*, volume 38, 2025. doi: 10.52202/085713-2102. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/5a9c1af5f76da0bd37903b6f23e96c74-Abstract-Conference.html.

[54] Runjin Chen, Andy Arditi, Henry Sleight, Owain Evans, and Jack Lindsey. Persona vectors: Monitoring and controlling character traits in language models, 2025. URL https://arxiv.org/abs/2507.21509v1. Version 1, 29 July 2025.

[55] Fan Zhou, Weitian Wang, and Tim Van de Cruys. Commitment before realization: When classifier-free guidance becomes unnecessary in masked diffusion language models, 2026. URL https://arxiv.org/abs/2608.08082v1. Version 1, 8 August 2026.

[56] Nicolas Leins, Jennifer Haase, Varvara Geronimus, Jana Gonnermann-Müller, and Sebastian Pokutta. Prompting against persona drift: Comparing intervention timing and content in LLM-simulated conversations, 2026. URL https://arxiv.org/abs/2609.24532v1. Version 1, 21 September 2026.

[57] Weinan E and Eric Vanden-Eijnden. Towards a theory of transition paths. *Journal of Statistical Physics*, 123(3):503–523, 2006. doi: 10.1007/s10955-005-9003-9.

[58] Weinan E and Eric Vanden-Eijnden. Transition-path theory and path-finding algorithms for the study of rare events. *Annual Review of Physical Chemistry*, 61(1):391–420, 2010. doi: 10.1146/annurev.physchem.040808.090412.

[59] Marten Scheffer, Jordi Bascompte, William A. Brock, Victor Brovkin, Stephen R. Carpenter, Vasilis Dakos, Hermann Held, Egbert H. van Nes, Max Rietkerk, and George Sugihara. Early-warning signals for critical transitions. *Nature*, 461(7260):53–59, 2009. doi: 10.1038/nature08227.

[60] H. A. Kramers. Brownian motion in a field of force and the diffusion model of chemical reactions. *Physica*, 7(4):284–304, 1940. doi: 10.1016/S0031-8914(40)90098-2.

[61] Matias I. Maturana, Christian Meisel, Katrina Dell, Philippa J. Karoly, Wendyl D'Souza, David B. Grayden, Anthony N. Burkitt, Premysl Jiruska, Jan Kudlacek, Jaroslav Hlinka, Mark J. Cook, Levin Kuhlmann, and Dean R. Freestone. Critical slowing down as a biomarker for seizure susceptibility. *Nature Communications*, 11:2172, 2020. doi: 10.1038/s41467-020-15908-3.

[62] Annika Hagemann, Jens Wilting, Bita Samimizad, Florian Mormann, and Viola Priesemann. Assessing criticality in pre-seizure single-neuron activity of human epileptic cortex. *PLOS Computational Biology*, 17(3):e1008773, 2021. doi: 10.1371/journal.pcbi.1008773.